

Formulas

Cartesian form

(y=x)

$$r = \frac{[1 + (y')^2]^{\frac{3}{2}}}{[y'']}$$

Parametric

x(t), y(t)

$$r = \frac{[(x')^2 + (y')^2]^{\frac{3}{2}}}{|x'y'' - y'x''|}$$

Polar

r, θ

$$r = \frac{[r^2 + (r')^2]^{\frac{3}{2}}}{|r^2 + 2(r')^2 - r r''|}$$

$$\textcircled{a}) \quad x = a \cos^3 \theta, \quad \theta = \frac{\pi}{6}$$

$$y = a \sin^3 \theta$$

$$x' = 4a \cos^2 \theta (-\sin \theta)$$

$$y' = 4a \sin^2 \theta (\cos \theta)$$

$$x'' = -4a \cos^2 \theta \sin \theta$$

$$y'' = 4a \sin^2 \theta \cos \theta$$

$$x''' =$$

$$r = \frac{[(x')^2 + (y')^2]^{\frac{3}{2}}}{|x'y'' - y'x''|}$$

$$\text{numerator} = [(-4a \cos^2 \theta \sin \theta)^2 + (4a \sin^2 \theta \cos \theta)^2]^{\frac{3}{2}}$$

$$= [(16a^2 \cos^4 \theta \sin^2 \theta) + (16a^2 \sin^4 \theta \cos^2 \theta)]^{\frac{3}{2}}$$

$$= [16a^2 \sin^2 \theta \cos^2 \theta (\sin^4 \theta + \cos^4 \theta)]^{\frac{3}{2}}$$

$$= 64a^3 \sin^3 \theta \cos^3 \theta (\sin^4 \theta + \cos^4 \theta)$$

Denominator =

This is way too much
I aint gonna do this
good luck to whoever who
completes this
bair bair

$$\theta) \quad y = c \cosh\left(\frac{x}{c}\right)$$

$$y' = c \sinh\left(\frac{x}{c}\right) \times \frac{1}{c}$$

$$y'' = \cosh\left(\frac{x}{c}\right) \times \frac{1}{c}$$

$$r = \frac{\left[1 + (y')^2\right]^{\frac{3}{2}}}{[y'']}$$

$$r = \frac{\left[1 + \left(\sinh\left(\frac{x}{c}\right)\right)^2\right]^{\frac{3}{2}}}{\left[c \cosh\left(\frac{x}{c}\right)\right]}$$

$$r = \frac{\left[1 + \sinh^2\left(\frac{x}{c}\right)\right]^{\frac{3}{2}}}{\frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

$$r = \frac{\left[1 + \sinh^2\left(\frac{x}{c}\right)\right]^{\frac{3}{2}}}{\frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

$$\therefore \cosh^2 u - \sinh^2 u = 1$$

$$1 + \sinh^2 u = \cosh^2 u$$

$$r = \frac{\left(\cosh^2 u\right)^{\frac{3}{2}}}{\frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

$$r = c \times \frac{\cosh^{\frac{3}{2}}\left(\frac{x}{c}\right)}{\cosh\left(\frac{x}{c}\right)}$$

$$r = c \times \cosh^2\left(\frac{x}{c}\right)$$

Qno 2)

$$x = a (\cos t + t \sin t)$$

$$y = a (\sin t - t \cos t) \quad a > 0$$

$$r = \left[(x')^2 + (y')^2 \right]^{3/2}$$

$$|x'y'' - y'x''|$$

$$\text{numerator} = \left[(a t \cos t)^2 + (a t \sin t)^2 \right]^{3/2}$$

$$= [a^2 t^2 \cos^2 t + a^2 t^2 \sin^2 t]^{3/2}$$

$$= [a^2 t^2 (\cos^2 t + \sin^2 t)]^{3/2} \quad \because \sin^2 t + \cos^2 t = 1$$

$$= (a^2 t^2)^{3/2}$$

$$= a^3 t^3$$

$$r = \frac{a^3 t^3}{a^2 t}$$

$$r = a^2 t$$

$$(uv)' = uv' + vu'$$

$$x = a (\cos t + t \sin t)$$

$$y = a (\sin t - t \cos t)$$

$$x' = a (-\sin t + \cos t + t \cos t)$$

$$y' = a (\cos t + \sin t - \cancel{\cos t})$$

$$x'' = a (-\cos t + \cos t)$$

$$y'' = a (-\sin t + \sin t)$$

$$\text{Denominator} = |(\cos t)(\cos t + \sin t) - (\sin t)(-\sin t + \cos t)|$$

$$= |a [\cos t (\cos t + \sin t)] - a [\sin t (-\sin t + \cos t)]|$$

$$= |a [\cos^2 t + \sin t \cos t] - a [-\sin^2 t + \sin t \cos t]|$$

$$= |a [\cos^2 t + \cancel{\sin t \cos t} + \sin^2 t - \cancel{\sin t \cos t}]|$$

$$= a [\cos^2 t + \sin^2 t]$$

$$= a$$

Q3)

distance r θ $\approx$ constant

$$r = \frac{[r^2 + (r')^2]^{3/2}}{|r^2 + 2(r')^2 - r r''|}$$

$$\begin{array}{l|l} r = \frac{a}{\theta} & r'' = -2(-1)a\theta^{-3} \\ r' = a \cdot \theta^{-1} & r'' = \frac{2a}{\theta^3} \\ r' = a(-1)\theta^{-2} & \\ r' = \frac{-a}{\theta^2} & \end{array}$$

Numerator = $[(\frac{a}{\theta})^2 + (-\frac{a}{\theta^2})^2]^{3/2}$

$$= \left[\frac{a^2}{\theta^2} + \frac{a^2}{\theta^4} \right]^{3/2}$$

=

$$= \left[\frac{a^2 \theta^2 + a^2 \theta^0}{\theta^4} \right]^{3/2}$$

$$= \left[\frac{a^2 \cancel{\theta^2} (\theta^2 + 1)}{\theta^4} \right]^{3/2}$$

$$= \frac{a^2}{\theta^4} (\theta^2 + 1)$$

$$= \frac{a^3}{\theta^6} (\theta^2 + 1)^{3/2}$$

Denominator = $1 (\frac{a}{\theta})^2 + 2 (-\frac{a}{\theta^2})^2 - (-\frac{a}{\theta}) (\frac{2a}{\theta^3})$

$$= 1 \frac{a^2}{\theta^2} - 2 \frac{a^2}{\theta^4} + \frac{2a^2}{\theta^4} 1$$

$$= 1 \frac{a^2}{\theta^2} - \frac{2a^2}{\theta^4} + \frac{2a^2}{\theta^4} 1$$

$$= 1 \frac{a^2}{\theta^2} 1$$

$$= \frac{a^2}{\theta^2}$$

$$r = \frac{a^3}{\theta^6} (\theta^2 + 1)^{3/2} \div \frac{a^2}{\theta^2}$$

$$= \frac{a^3}{\theta^6} \times \frac{\theta^2}{\theta^2} \times (\theta^2 + 1)^{3/2}$$

$$= \frac{a}{\theta^4} (\theta^2 + 1)^{3/2}$$

$$b^1) \quad r = 2 \cos 2\theta$$

$$\theta = \frac{\pi}{12}$$

$$r = 2 \cos(2\theta)$$

$$r' = -4 \sin 2\theta$$

$$r' = -2 \sin 2\theta \cdot 2$$

$$r'' = -8 \cos 2\theta$$

$$r' = -4 \sin 2\theta$$

$$r = \frac{[r^2 + (r')^2]^{3/2}}{|r^2 + 2(r')^2 - rr''|}$$

numerator =

$\theta) \quad r = a(1 + \cos \theta) \quad \text{prove} \quad \frac{1}{r^2} = \frac{8a}{9}$

$$r = a(1 + \cos \theta) \quad | \quad r'' = -a(\cos \theta)$$

$$r' = a(0 - \sin \theta)$$

$$r' = -a(\sin \theta)$$

$$r = [r^2 + (r')^2]^{1/2}$$

$$[r^2 + 2(r')^2 - r r'']$$

$$r = [a^2(1 + \cos \theta)^2 + (-a \sin \theta)^2]^{1/2}$$

$$a^2(1 + \cos \theta)^2 + 2(-a \sin \theta)^2 - (a(1 + \cos \theta))(-a \cos \theta)$$

$$r = [a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta]^{1/2}$$

$$a^2(1 + \cos \theta)^2 + 2a^2 \sin^2 \theta + a \cos \theta (a + a \cos \theta)$$

$$r = [a^2(1^2 + 2\cos \theta + \cos^2 \theta) + a^2 \sin^2 \theta]^{1/2}$$

$$a^2(1^2 + 2\cos \theta + \cos^2 \theta) + 2a^2 \sin^2 \theta + a^2 \cos \theta + a^2 \cos^2 \theta$$

$$r = [a^2(1 + 2\cos \theta + \cos^2 \theta + \sin^2 \theta)]^{1/2}$$

$$a^2(1 + 2\cos \theta + \cos^2 \theta + \sin^2 \theta)$$

$$r = [a^2(1 + 2\cos \theta)]^{1/2}$$

$$a^2$$

? do it again

$$\theta) \quad y = c \cosh\left(\frac{x}{c}\right)$$

$$r = \frac{\left[1 + (y')^2\right]^{\frac{3}{2}}}{[y'']}$$

$$= \frac{\left[1 + \left(\sinh\left(\frac{x}{c}\right)\right)^2\right]^{\frac{3}{2}}}{\frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

$$= \frac{\left[1 + \sinh^2\left(\frac{x}{c}\right)\right]^{\frac{3}{2}}}{\frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

$$= \frac{\left[\cosh^2\left(\frac{x}{c}\right)\right]^{\frac{3}{2}}}{\frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

$$= \frac{\cosh^3\left(\frac{x}{c}\right)}{\frac{1}{c} \cosh\left(\frac{x}{c}\right)}$$

$$= c \cdot \cosh^2\left(\frac{x}{c}\right)$$

$$y = c \cosh\left(\frac{x}{c}\right)$$

$$y' = \sinh\left(\frac{x}{c}\right)$$

$$y'' = \frac{1}{c} \cosh\left(\frac{x}{c}\right)$$

Qno 2)

$$x = a (\cos t + t \sin t)$$

$$y = a (\sin t - t \cos t)$$

$$r = \left[(x')^2 + (y')^2 \right]^{3/2}$$

$$|x'y'' - y'x''|$$

$$x' = a (-\sin t + (t \cdot \cos t + \sin t))$$

$$x' = a (-\cancel{\sin t} + t \cdot \cos t + \cancel{\sin t})$$

$$x' = a t \cos t$$

$$x'' = a (t(-\sin t) + \cos t)$$

$$x'' = a (-t \sin t + \cos t)$$

$$y' = a (\cos t - (t(-\sin t) + \cos t))$$

$$y' = a (\cos t + t \sin t - \cancel{\cos t})$$

$$y' = a t \sin t$$

$$y'' = a (t \cos t + \sin t)$$

$$y'' = a (t \cos t + \sin t)$$

$$\text{Num} = \left[(a t \cos t)^2 + (a t \sin t)^2 \right]^{3/2}$$

$$= \left[a^2 t^2 \cos^2 t + a^2 t^2 \sin^2 t \right]^{3/2}$$

$$= \left[a^2 t^2 (\sin^2 t + \cos^2 t) \right]^{3/2}$$

$$= \left[a^2 t^2 (1) \right]^{3/2}$$

$$= (a^2 t^2)^{3/2}$$

$$= a^3 t^3$$

$$\text{Deno} = | a t \cos t (a t \cos t + a \sin t) - a t \sin t (-a \sin t + a \cos t) |$$

$$= | a^2 t^2 \cos^2 t + a^2 t \cancel{\sin t} \cos t + a^2 t^2 \sin^2 t - a^2 t \cancel{\sin t} \cos t |$$

$$= | a^2 t^2 (\sin^2 t + \cos^2 t) |$$

$$= | a^2 t^2 |$$

$$= a^2 t^2$$

$$= \frac{a^3 t^3}{a^2 t^2}$$

$$= a t$$

Q3)

r — distance
 θ — angle
 $= a$ — constant

$$r = \frac{a}{\theta}$$

$$r = a \theta^{-1}$$

$$r' = a(-1) \theta^{-2}$$

$$r' = -\frac{a}{\theta^2}$$

$$r'' = -a \theta^{-3}$$

$$r'' = -a(-2) \theta^{-4}$$

$$r'' = \frac{2a}{\theta^4}$$

$$r = \frac{[r^2 + (r')^2]^{3/2}}{|r^2 + 2(r')^2 - r r''|}$$

$$\text{Numerator} = \left[\left(\frac{a}{\theta} \right)^2 + \left(-\frac{a}{\theta^2} \right)^2 \right]^{3/2}$$

$$= \left[\frac{a^2}{\theta^2} + \frac{a^2}{\theta^4} \right]^{3/2}$$

$$= \left[\frac{a^2 \theta^2 + a^2}{\theta^4} \right]^{3/2}$$

$$= \left[\frac{a^2}{\theta^4} (\theta^2 + 1) \right]^{3/2}$$

$$= \frac{a^3}{\theta^6} (\theta^2 + 1)^{3/2}$$

$$\text{Denominator} = \left| \left(\frac{a}{\theta} \right)^2 + 2 \left(-\frac{a}{\theta^2} \right)^2 - \left(\frac{a}{\theta} \right) \left(\frac{2a}{\theta^4} \right) \right|$$

$$= \frac{a^2}{\theta^2} + \frac{2a^2}{\theta^4} - \frac{2a^2}{\theta^4}$$

$$= \frac{a^2}{\theta^2}$$

$$b^4) \quad r = 2 \cos 2\theta$$

$$\theta = \frac{\pi}{12}$$

$$r = \frac{[r^2 + (r')^2]^{\frac{3}{2}}}{|r^2 + 2(r')^2 - rr''|}$$

$$r = 2 \cos 2\theta$$

$$r' = -4 \sin 2\theta$$

$$r' = 2(2) - \sin 2\theta$$

$$r'' = -4(2) \cos 2\theta$$

$$r' = -4 \sin 2\theta$$

$$r'' = -8 \cos 2\theta$$

$$\text{num} = [(2 \cos 2\theta)^2 + (-4 \sin 2\theta)^2]^{\frac{3}{2}}$$

$$\text{Denominator} = | (2 \cos 2\theta)^2 + 2(-4 \sin 2\theta)^2 - (2 \cos 2\theta)(-8 \cos 2\theta) |$$

$$= [4 \cos^2 2\theta + 16 \sin^2 2\theta]^{\frac{3}{2}}$$

$$= | 4 \cos^2 2\theta + 2(16 \sin^2 2\theta) + 16 \cos^2 2\theta |$$

=

$$= 20 \cos^2 2\theta + 32 \sin^2 2\theta$$

$$= [4 \cos^2 2\theta + 16 \sin^2 2\theta]^{\frac{3}{2}}$$

$$= | 15 + 8 |$$

$$[3 + 4]^{\frac{3}{2}}$$

$$= \frac{(7)^{\frac{3}{2}}}{2^3}$$

$$= 0.865222$$